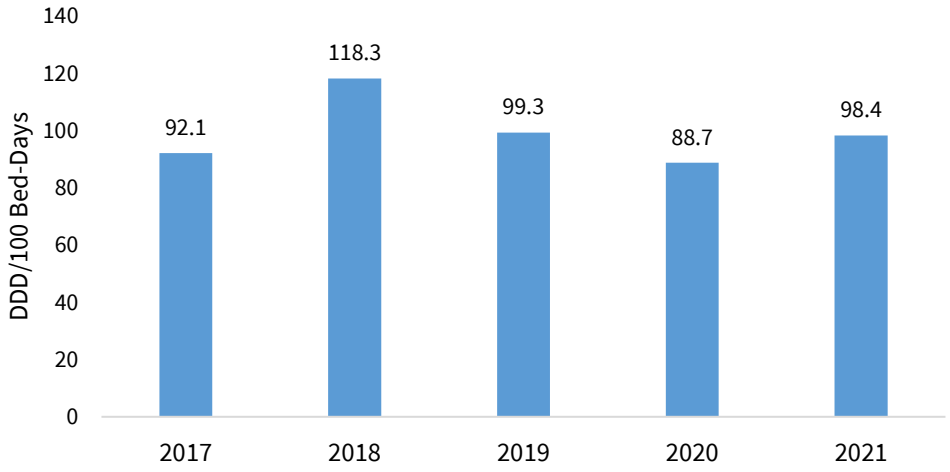
Results

Each of the NAC-NET sites had a different bed strength in the 5 years for which this analysis was done. A cumulative bed strength of each year for the 10 sites is given in Table 1.

Table 1: Year wise cumulative bed strength

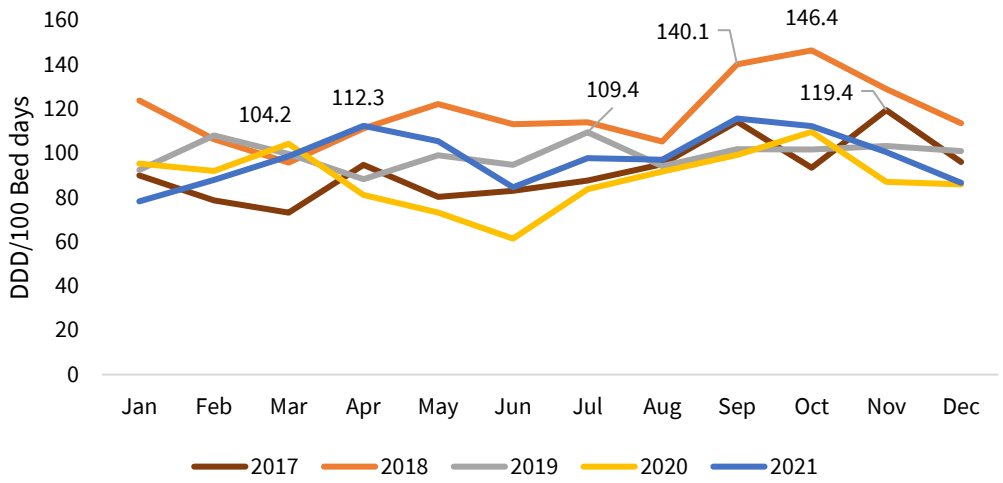
Year	Cumulative beds
2017	13838
2018	14088
2019	14889
2020	15168
2021	15532

Figure 3: Antibiotic consumption from 2017-2021



The antibiotic consumption across the ten sites ranged from 88.7 (2020) to 118.3 (2018) DDD/100-bed-days over the five years.

Figure 4: Trend of antibiotic consumption from 2017-2021



The trend of antibiotic consumption shows a surge in November 2017, September-October 2018, July 2019 and April 2021.

AWaRe classification of antibiotics

The antibiotics used in the hospitals are classified based on WHO AWaRe classification into three groups (8)

- Access group of antibiotics have activity against a wide range of commonly encountered susceptible pathogens while also showing lower resistance potential than antibiotics in the other groups
- Watch group of antibiotics have higher resistance potential and includes most of the highest priority agents
- Reserve group of antibiotics includes antibiotics and antibiotic classes that should be reserved for treatment of confirmed or suspected infections due to multi-drug-resistant organisms.

Figure 5: Consumption of antibiotics as per WHO AWaRe classification (2017-2021)

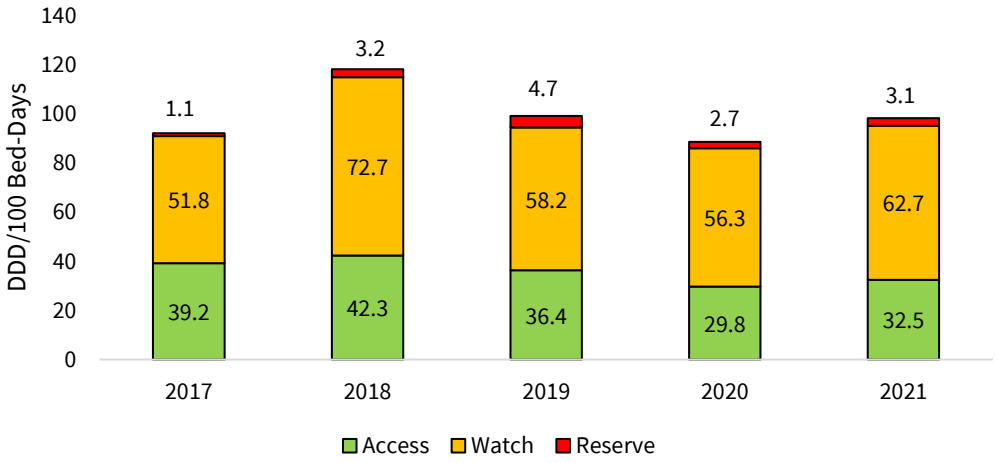
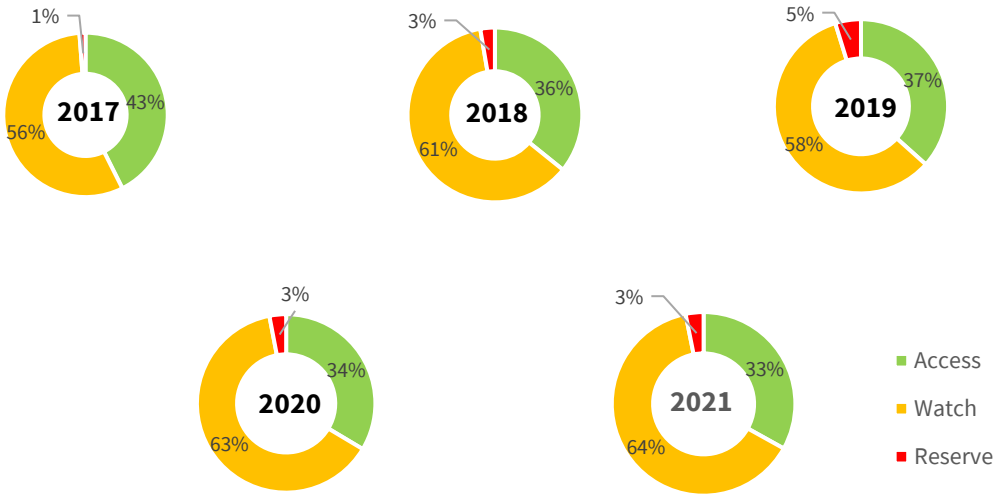


Figure 6: Proportion of antibiotic consumption based on WHO AWaRe classification (2017-2021)



Access group consumption ranged from 29.8 to 42.3 DDD per 100-bed-days with 2017 recording the highest proportion of Access group consumption at 43%. The Watch group consumption steadily increased from 51.8 to 72.7 DDD/100-bed-days with 2021 recording the highest proportion of Watch group consumption at 64%. Reserve group consumption ranged from 1.1 to 4.7 DDD/100-bed-days with the highest consumption seen in 2019.

Figure 7: Trend of antibiotic consumption as per WHO AWaRe classification (2017-2021)

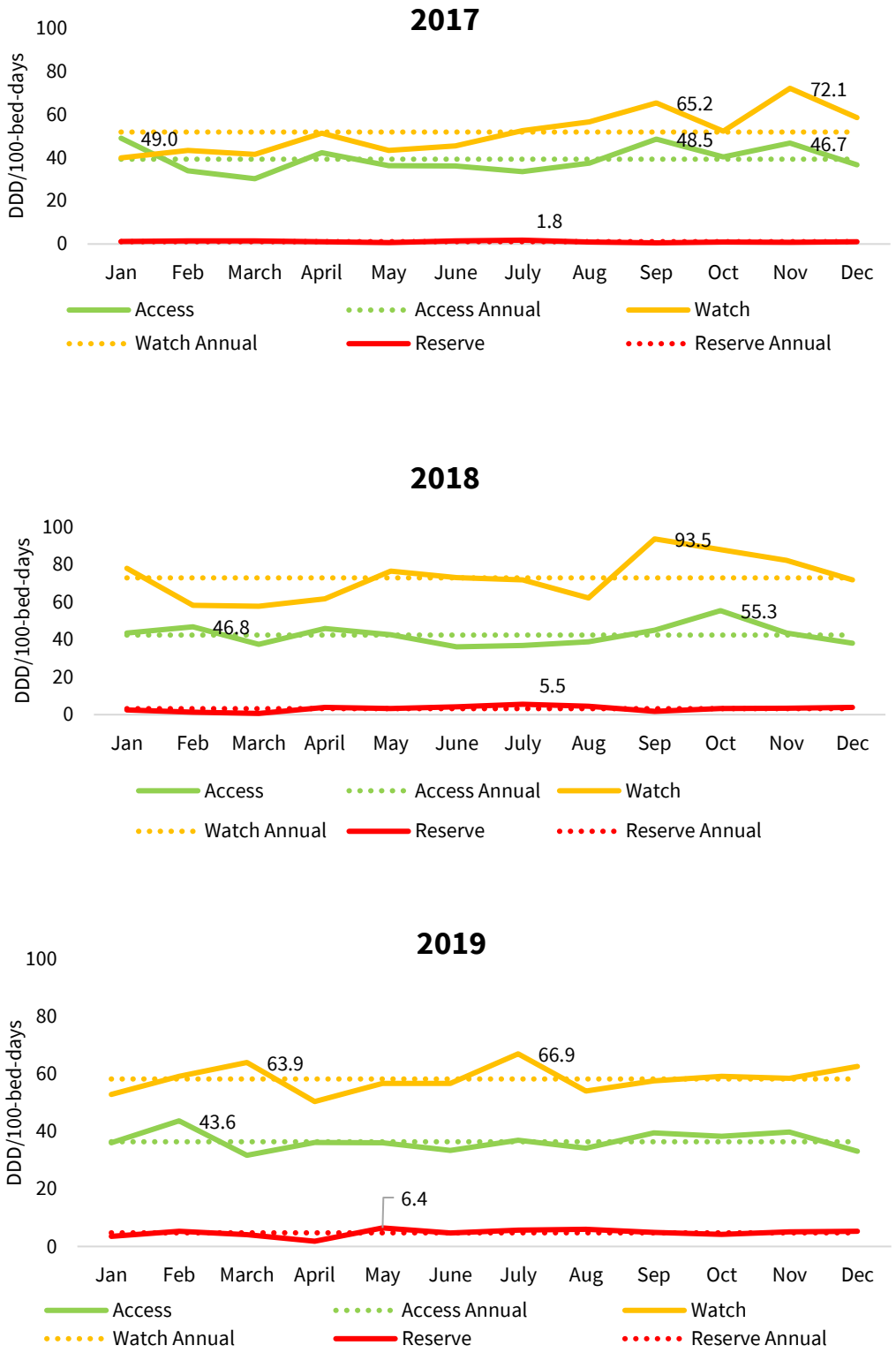


Figure 10: Year wise top 10 antibiotics from 2017-2021

